

The scalar character of the unitization

`MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.infinityCharacter`

def

Names the character which reads the scalar coordinate and annihilates the embedded original algebra.

Statement

Let A be a complex C^* -algebra, with no unit assumed. Write its complex unitization as $\tilde{A} = \mathbb{C} \oplus A$, with elements (λ, a) and inclusion $\iota(a) = (0, a)$. The scalar character is a character of $\tilde{A}$.

Definition

Multiplication in the unitization is given by $(\lambda, a)(\mu, b) = (\lambda\mu, \lambda b + \mu a + ab)$. Thus

$$\chi_\infty : \tilde{A} \rightarrow \mathbb{C}, \quad \chi_\infty(\lambda, a) = \lambda$$

is a complex algebra homomorphism with $\chi_\infty(1, 0) = 1$, hence a nonzero character. The RHS below presents this very scalar-coordinate homomorphism as an element of the character space.

Assumptions

There is no nontriviality, order, separability, or representation-space hypothesis in this definition.

Conclusion

The resulting character satisfies $\chi_\infty(\lambda, a) = \lambda$, and consequently $\chi_\infty(\iota(a)) = 0$ for every $a \in A$.

For a closed unital $*$ -subalgebra $D \subseteq \tilde{A}$, the separate definition `infinityCharacterOn D` is its restriction $\chi_\infty|_D$. It is not a new character of A .

Main citations

- Exact definition and context

Lean source signature (exact)

The complete declaration below is a separate exact source excerpt; the original header record is retained with the manuscript.

```
-- The scalar character of the minimal unitization. -/
noncomputable def infinityCharacter :
  WeakDual.characterSpace ℂ (Unitization ℂ A) :=
  WeakDual.CharacterSpace.equivAlgHom.symm (Unitization.fstHom (R := ℂ) (A := A))
```

In the source	Mathematical meaning
Unitization $\mathbb{C} \oplus A$	The complex unitization $\tilde{A} = \mathbb{C} \oplus A$, not the original algebra alone.
WeakDual.characterSpace $\mathbb{C} \times (\text{Unitization } \mathbb{C} \oplus A)$	The output belongs to the space of nonzero continuous complex characters of $\tilde{A}$.
Unitization.fstHom $(R := \mathbb{C}) (A := A)$	The algebra homomorphism $(\lambda, a) \mapsto \lambda$.
WeakDual.CharacterSpace.equivAlgHom.symm	Express that same homomorphism as a character; its value on each (λ, a) remains λ .

Exact source and proof.

Earlier published Card and PDF

▼ Exact content identity

Declaration: MathlibAnnex.Analysis.CStarAlgebra.NonUnitalCStarRepresentation.infinityCharacter
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